

Mohammad Mahmudul Hasan

Computer vision and LLMs | Team Lead, AI Engineer

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RESEARCH PROFILE

AI engineering team lead with applied computer-vision experience in road imagery, building-change analysis, point-cloud interpretation and a co-authored explainable medical-imaging paper. Seeking research in robust visual understanding, vision-language models and evidence-grounded LLM systems.

EDUCATION

Islamic University of Technology, Bangladesh | BSc in Electrical and Electronic Engineering, completed 22 March 2021.

CGPA 3.76/4.00; first class with honours. Project and Thesis: A+; Digital Signal Processing: A; Artificial Neural Networks and Fuzzy Logic: A+.

PROFESSIONAL EXPERIENCE

HawarIT Limited — Team Lead, AI Engineer | with the company since 2021; team-lead role since [ADD month and year of promotion].

Currently lead a six-member AI engineering team.

- Worked with car-captured 360-degree imagery to support traffic-sign condition assessment and missing-asset review.
- Worked on multi-year orthophoto building-change analysis and point-cloud segmentation/classification.
- Currently lead a six-member AI engineering team; combine production visual-data experience with a co-authored explainable medical-imaging publication.

Earlier titles, exact dates and individual project ownership: [ADD verified career chronology].

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, Mohammad Mahmudul Hasan, Abrar Islam, and Md. Golam Rabiul Alam. “Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI.” 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2021. DOI: [10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

Individual contribution: [ADD precisely what you did]. The author preprint is the same work.

SELECTED RESEARCH AND TECHNICAL WORK

TRAFFIC-SIGN REVIEW FROM 360-DEGREE IMAGERY | Professional experience

Worked with car-captured 360-degree imagery to identify faulty or missing traffic signs and review their quality.

Research question: Can a visual system distinguish a genuine defect from an uncertain observation and communicate the evidence to a reviewer?

MULTI-CLASS TRAFFIC DETECTION | Competition and public code

The earlier CV describes data cleaning, preprocessing, augmentation and comparison of YOLOv3, YOLOv5 and R-CNN approaches for 21-class traffic detection. A public repository describes detection of 21 traffic labels.

Research question: How does class imbalance or scene shift affect a detector, and what evidence is needed to compare models fairly?

EXPLAINABLE RETINAL OCT CLASSIFICATION | Peer-reviewed conference publication

Co-authored Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI. The study examines a compact CNN and LIME-based explanations.

Research question: How stable are explanations under image perturbations, and how should they be evaluated alongside classification and calibration?

RESEARCH DIRECTION

Can multimodal systems produce useful inspection reports while keeping every factual claim traceable to image or document evidence? Develop a small benchmark of visual observations and reference-grounded questions. Compare a detector-plus-rules baseline, a vision-language model and a retrieval-assisted system; measure factual support, missed defects, abstention and cost.

TECHNICAL FOUNDATION

Computer vision; image analysis; point-cloud classification; data preparation; Python; MATLAB. Current ML frameworks and deployment stack: [ADD verified tools]. LLM/RAG/fine-tuning experience: [ADD only completed, evidenced work].

ADDITIONAL CONTEXT

Independent AI/ML study during the final semesters of the EEE degree and participation in AI competitions. Historical Coursera credentials and competition placement should be added with year and verification links.

Language: Bangla (native); English-medium instruction certificate available. IELTS pending; intended overall score 7.0 is not an achieved score.

References: [ADD academic and professional referees as required by the program].